



ERC standard anatomical knee reference frames.

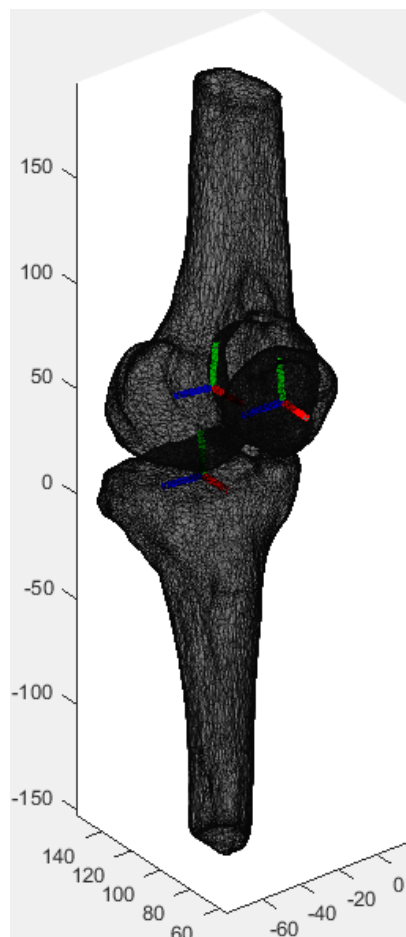


Figure 0: Resulting anatomical reference frames.

Knee reference frames

This document comes with version 1.1 of the ERCKneeReferenceFrames code. This program calculates the reference frames for (any of) the three major knee-bones, and is designed to work with partial femura and tibia. For constraints on the surface meshes this program can process see the next section. The short definitions of the reference frames are given after that, which are followed by a thorough explanation of the procedures, illustrated using figures generated using the different DiagInfo plotting functions available in the source (see the demo). After that short descriptions are given of some experiments performed to determine the parameter values. Verifying experiments for the different steps are described in the respective text.

If you want to know how to use the functions, you should use matlabs help function (e.g. 'help ERCKneeReferenceFrames'), or see the demo (demo.m) for a quick overview of the functionality of the main function called ERCKneeReferenceFrames.

Contents

Knee reference frames.....	2
Notes on using the ERCKneeReferenceFrames.....	3
Program structure.....	3
Short Definition of the reference frames	4
Detailed procedure descriptions.....	5
Determining inertial axes.....	5
Calculating the cross section area	5
Reference frame estimate	5
Other external frame estimates.....	6
Femoral reference length	7
Verifying the reference length.....	7
Detailed reference frame procedure descriptions	8
Patellar reference frame.....	8
Finding the inferior pole	9
Tibial reference frame.....	10
Femoral reference frame	11
Verification experiment	17
Conclusions	17
Qualitative results.....	18
Open questions	19
Changes from the Lianne code	20
References	21

Notes on using the ERCKneeReferenceFrames

This code puts some loose constraints and some preferences on the data:

- Make sure that all stl files are in millimeters. Some parameters are in this unit.
- All meshes must be triangular meshes.
- When ERCKneeReferenceFrames loads the stl files from the paths, it assumes they are binary (not ascii). Non-binary stl files have to be loaded externally, and passed to ERCKneeReferenceFrames as a struct with the fields F and V, holding the faces and vertices.
- Relatively uniform surface vertex densities are recommended. Sometimes the mean vertex is used as a measure of the center of mass, and other times “thin” slices are taken from the stl. Both these steps benefit from uniform vertex densities.
- Meshes should be closed (no holes). Inertial axis estimation uses a flood filling algorithm that fills up to the boundary of the stl. Holes in the stl break this (look for the “volume fully flooded” warning). Also cutting stls and closing the surface relies on no holes to identify wrong, exterior faces (i.e. in a concave area).
- Limit the number of vertices to something reasonable. If MATLAB runs into memory problems, decrease the vertex count.
- A femur length of 18cm guarantees reproducible results. An anatomical reference length is calculated from the morphology of the femur’s condyles. The longest length required in the verification dataset was 16.5 cm.
- When generating the reference frames for multiple bones with one call, make sure they are all registered to the same global space. This is assumed by, among others, the external frame estimation using all bones.
- Relatively smooth surfaces improve the quality of the result. The inferior pole finding algorithm finds some extreme vertex to be designated as the inferior pole. The ML axis for the femur depends on cylinder fits to the articulating surfaces, which are, among others, identified by their vertex normals. Meshes resembling voxel meshes strongly impair both these procedures.

Program structure

The main function an end user will be using is ERCKneeReferenceFrames, which does precutting and external frame estimation, before calling starting the routines for the separate bones (ERCrefFrame(...)). Notable exceptions are the three functions called plot<BONE>DiagInfo, where <BONE> is either Femur, Patella or Tibia. These functions are used to visualize the intermediate steps of the different procedures (see the demo for an example). All figures accompanying the detailed descriptions of the algorithm found later in this document are generated using those functions.

Short Definition of the reference frames

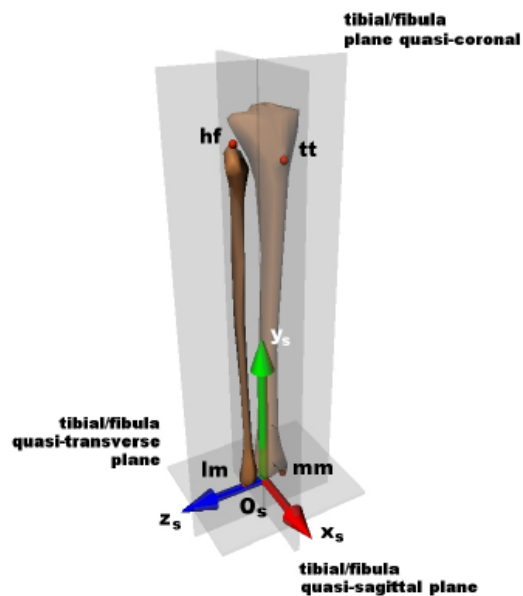
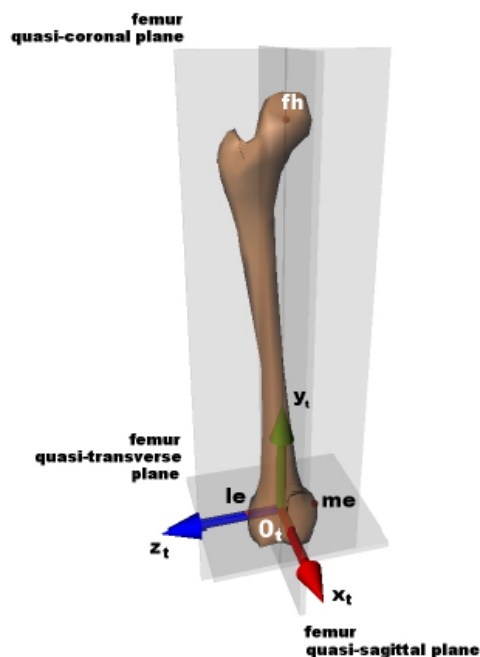


figure 4. Tibial/Fibula anatomical frame.

Figure 1: Reference frame convention. X Y and Z correspond to AP, PD and ML respectively. X points forwards, Y points up and Z is such that we have a right handed reference frame.

We adhere to the reference frame convention as advised by the European society for biomechanics in so far as that our definitions of X Y and Z correspond to the AP, PD and ML axes respectively, together forming a right handed reference frames. Here we give the short definitions of the frames, but keep in mind that in order to reproducibly generate the reference frames the whole procedure is necessary.

Femur

Z: The line connecting the centers of two cylinder fits to the articulating surfaces

X: The normal to the plane containing the Z axis and smallest diaphysis inertial axis (along the length).

Y: In said plane, and perpendicular to Z. (consequently almost the diaphysis inertial axis)

Tibia

All inertial axes of the tibial plateau. From smallest to largest: Z X Y.

Patella

X: Normal to the plane fitting the anterior surface

Z: Normal to the plane holding the X axis and the line connecting the center of mass to the inferior pole

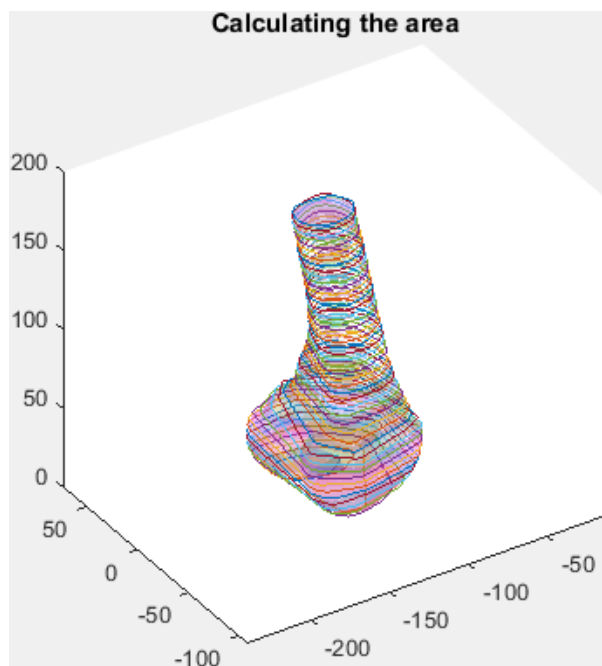
Y: In said plane, perpendicular to both X and Z. (consequently similar to the line connecting the CM to the inferior pole)

Detailed procedure descriptions

The procedure to generate the reference frames is different for the three bones, but some elements are reoccurring. These are explained in this section.

Determining inertial axes

Determining the inertial axes of the different bones and sections is done on the basis of point-masses, distributed uniformly on a square grid throughout the volume (using a flood filling algorithm after filling voxels on the surface). It turns out that the grid size is an important parameter determining the reproducibility of the different reference frames. Different grid sizes have been chosen for the different bones and steps, optimized for both reproducibility and runtime.



Calculating the cross section area

Calculating the cross sectional area is done by rotating the vertices of the mesh such that the direction of the cross sectional area is one of the major axes. Subsequently the vertices are separated in that direction in equal sized parts on the basis of this coordinate (using a histogramming procedure), and a 2D convex hull is calculated for each section. Note that since we're using a convex hull, it is not necessarily the cross section area we are measuring, it also includes for instance the empty space between the articulating surfaces of the femur.

Figure 2: Calculating the cross-section area (CA) for a distal femur. First the bone is separated into equal sized sections (e.g. 2%). The cross-section

area is determined by the area of a 2D convex hull around the points of a section. This results in an overestimate of the area when it includes empty spaces (e.g. in between the articulating surfaces).

Reference frame estimate

The procedures described and implemented for the different bones are able to generate the directions of the reference axes, but not their sign (i.e. determine PD, but not which is P). This is solved by a reference frame that is used to determine the signs, and as an initial guess for some procedures. Several ways to determine this estimate are available. For its usage to determine the signs it has to be correct within a margin of about 90 degrees, which means that any¹ reference frame could work as long as the signs are right. Because it is also used as an initial guess for the inferior pole finding, and used to align intermediate solutions, it is advised try to estimate the axes within a ± 45 degree accuracy.

¹ Regardless of its orientation

External reference frame estimation

An external reference frame (see figure 1) can be defined on the basis of the centers of masses (CM) of the 3 different bones, estimated by the mean vertex for a mesh. The rationales here are that the tibia is more distal than the femur and the patella is more anterior than both.

In this reference frame estimate the proximal-distal axis (PD / Y) goes from the femur's CM to the tibial CM (in red). The anterior-posterior axis (AP, in blue / Z) is in a direction perpendicular to the PD axis, connecting that axis to the patellar CM (angle between the red and blue long lines are 90deg). The medio-lateral axis (ML) is perpendicular to the AP and PD axes, and is positive towards the right handed side. It is determined by taking the cross product $ML = AP \times PD$.

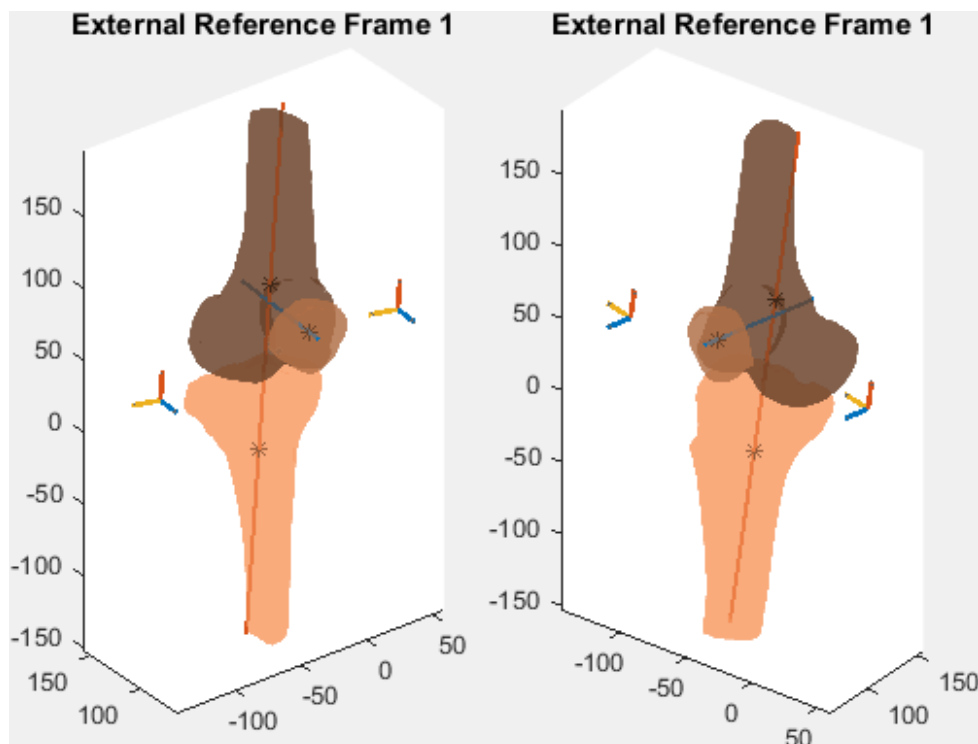


Fig 3: The external reference frame. First the three bone centers (mean vertex) are determined. The external PD axis connects the femur and tibial centers. The AP axis is perpendicular to the PD axis, and goes through both the PD axis and the patellar center. The ML axis is chosen such that we produce an orthonormal, right handed reference frame. The two panels show the same object from a different angle.

Other external frame estimates

When not all bones are present, other ways have to be employed to estimate the directions of the axes. Version 1.0 is able to estimate the PD axis from the shape of the femur. The other direction, or when the femur is not present, have to be selected manually in the UI's that pop up when running ERCKneeReferenceFrames.m.

Femoral reference length

The diaphysis is not straight, but has a curve. This means that the long inertial axis direction changes when more femur is used to determine the reference frame. To standardize this in a way that can be reconstructed from the distal femur alone, we calculate the reference length as the average side length of a 2D bounding box around the vertices of the epicondyles, identified by the first principle component of the surface vertices. Twice the reference length is used to determine the femoral reference frame, which results in a minimal distal femur length of about 13 to 17 cm.

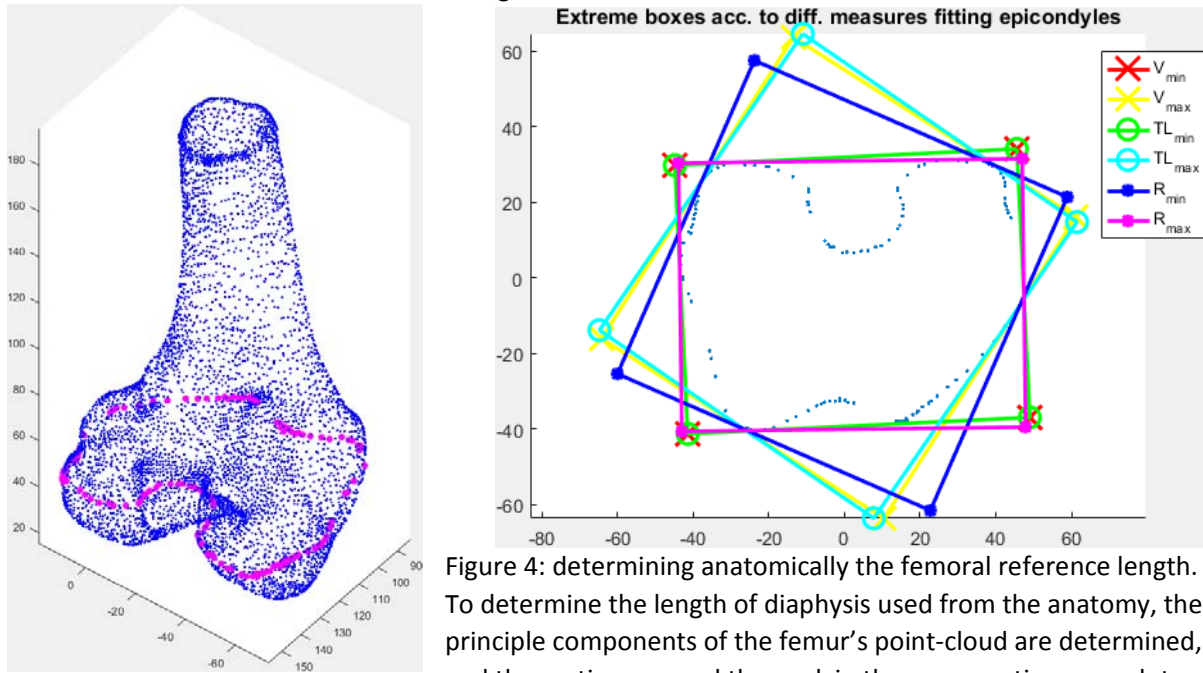


Figure 4: determining anatomically the femoral reference length. To determine the length of diaphysis used from the anatomy, the principle components of the femur's point-cloud are determined, and the vertices around the peak in the cross-section area plot are

selected (pink dots left panel). Different boxes have been investigated, we decide to use the minimal area rectangle ($V_{\min}$), and take the mean side length as the reference length.

Verifying the reference length

Because the reference length procedure is novel, experiments have been performed to evaluate its robustness to input-femur length, as well as the dependence of the resulting frame on the used femoral length.

The robustness has been investigated using the whole dataset of 40 bones. For those bones the reference length was calculated starting from the whole source mesh, and cutting 10% of each step, until less than the reference length remains. Whole bones were first cut in half. Comparing the minimal and maximal value from those (usually about 5), result in a worst case difference of 5% comparing, with a peak at 1.5%. The conclusion is that the reference length calculation is very consistent when using different lengths of input femur.

In another experiment aimed at determining the appropriate factor on the length (we choose 2) 9 femurs were processed, using increasingly less diaphysis length. The factor of 2 was determined to be the appropriate tradeoff by finding that the deviation from the result produced by the whole mesh is rather small, but we do not require too much bone. It turns out that about 13 to 17 cm is usually required.

Detailed reference frame procedure descriptions

This sections contain in-depth descriptions of the procedures to derive the different reference frames. The accompanying figures illustrates all the steps, and together with their captions can probably serve as a sufficient explanation for most users. These figures are generated with their respective `plotXxxDiagInfo`, which are useful to diagnose unexpected results.

Patellar reference frame

The patellar reference frame definition and determination are based on the works of Ho[5] and Sharma[6]. First, the patellar CM and inertial axes are determined. Subsequently the anterior patellar surface is separated from the patella, with the AP direction given by the largest inertial axis. The vertices in the most anterior 30% of space are selected and a plane is fitted (least squared) to them (notably (usually) not 30% of the vertices furthest anterior). The normal to the plane fitted to the anterior patella defines the AP axis (X, red in Fig5).

In the next step the inferior pole is determines (see below). The proximal-distal axis is similar to the line connecting the patellar center of mass to the inferior pole (green line), but perpendicular to the AP axis.

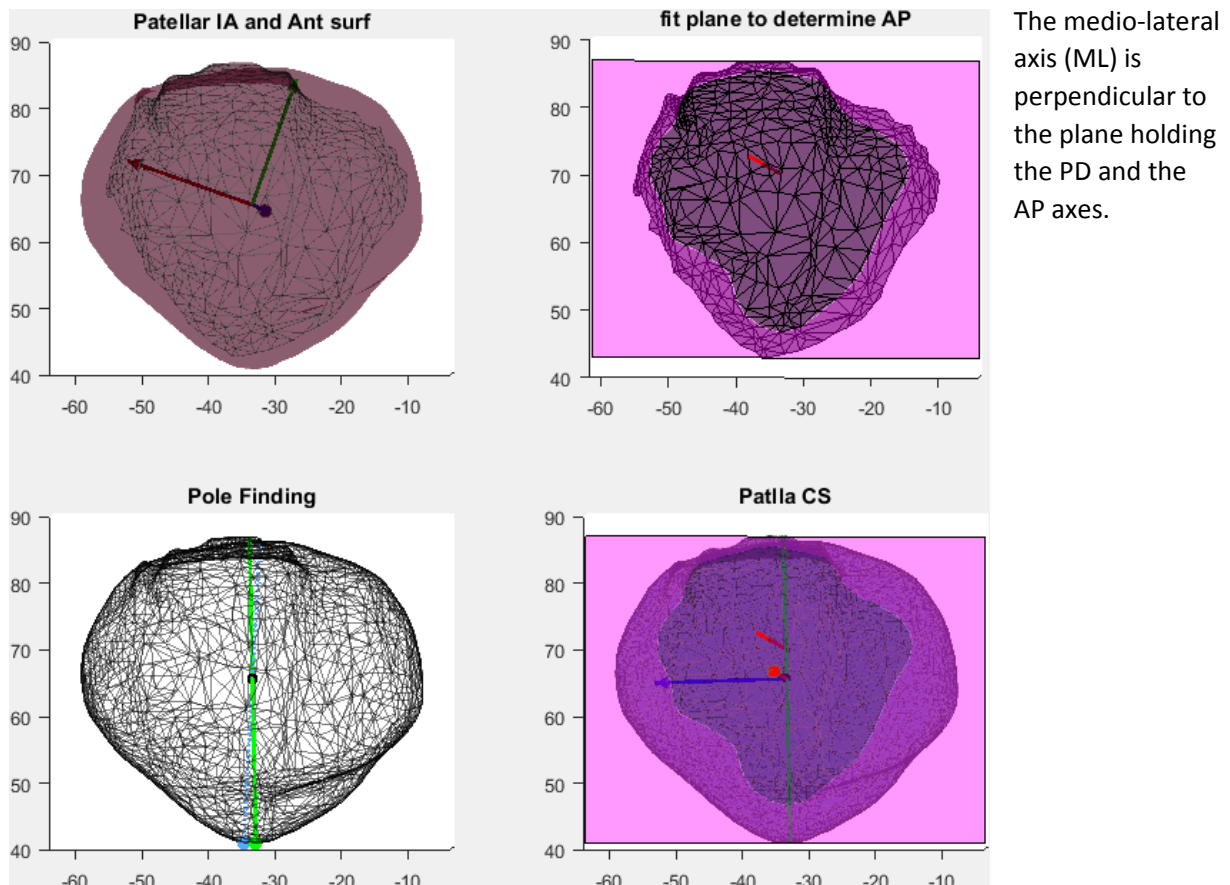


Figure 5: Output of `plotPatellaDiagInfo.m(1/2)`: Patellar Reference frame determination. Patellar inertial axis, separated posterior surface, and eventual coordinate system. Note the two points (blue and green) around the inferior pole in panel 3, which are the results of the 1st and 2nd (and 3rd, but this overlaps) iteration of the pole finding algorithm.

Finding the inferior pole

Finding the inferior pole is done by a custom algorithm verified on the TLEM dataset. The procedure is as follows.

Until you find the same vertex twice:

- Stretch the patella in the direction you think the pole is in by a certain factor (1.5).
- Determine the best guess pole as the vertex furthest away from the center of mass in the scaled patella.
- Update the best guess direction to be toward the newly found best guess

For the first guess we use PD axis of the reference frame estimate. The inferior pole is found robustly for all patella it is tested on, as long as their meshes are reasonable (i.e. do not have a 2cm wide flat surface inferiorly...).

The rationale behind the scaling and distance measure is that the inferior pole is far away from the center of mass of the patella, but it is also distally from the CM. The iterations reduce the influence of the first guess (external axis), and make sure the pole is found at the tip of some pointy structure.

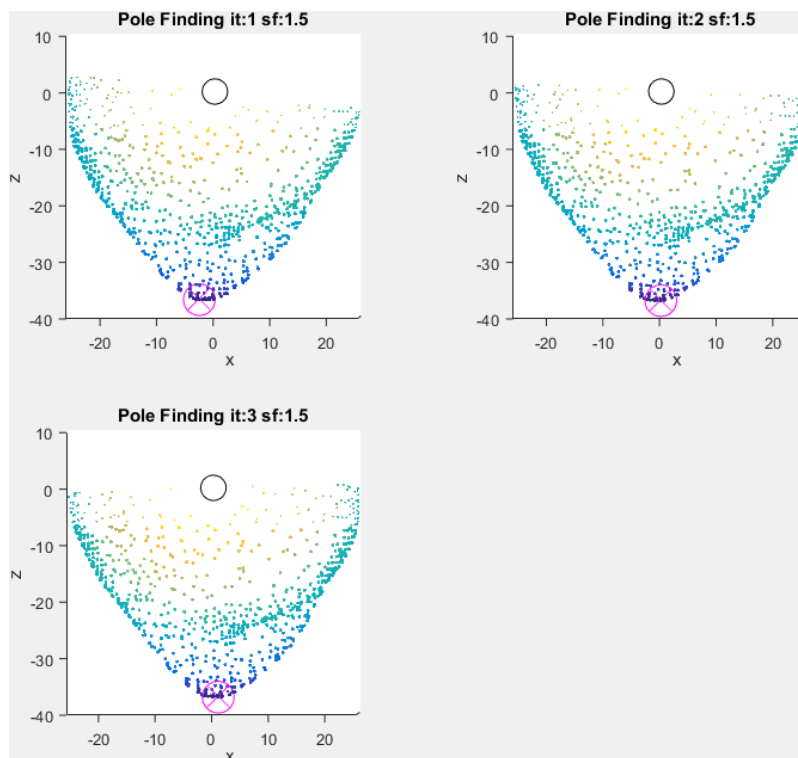


Figure 6: Output of plotPatellaDiagInfo.m (2/2): Determining the inferior pole using a scaling measure and guess direction. The colors represent the value of the distance measure. In each iteration the patella is stretched 1.5 times towards the current guess direction, and the furthest vertex (pink crosshairs) from the center of mass (black circle) is found and taken as the next guess direction. This is repeated until the same vertex is found twice. Note that we are looking at 3D point clouds.

Tibial reference frame

The first steps for the tibial reference frame are the same as for the patella, the center of mass and inertial axes are determined. Along the smallest inertial axis (length of the bone), the cross sectional area is determined. The plane perpendicular to this axis at the point of maximum cross-sectional area is used to separate the tibial plateau from the shaft. The center of mass and inertial axes of the plateau are determined after separation. These give the reference frame for the tibia:

The largest inertial axis of the tibial plateau corresponds to the tibial proximal-distal axis (PD), while the smallest is ML and the last one gives AP. The origin is determined as the tibial CM.

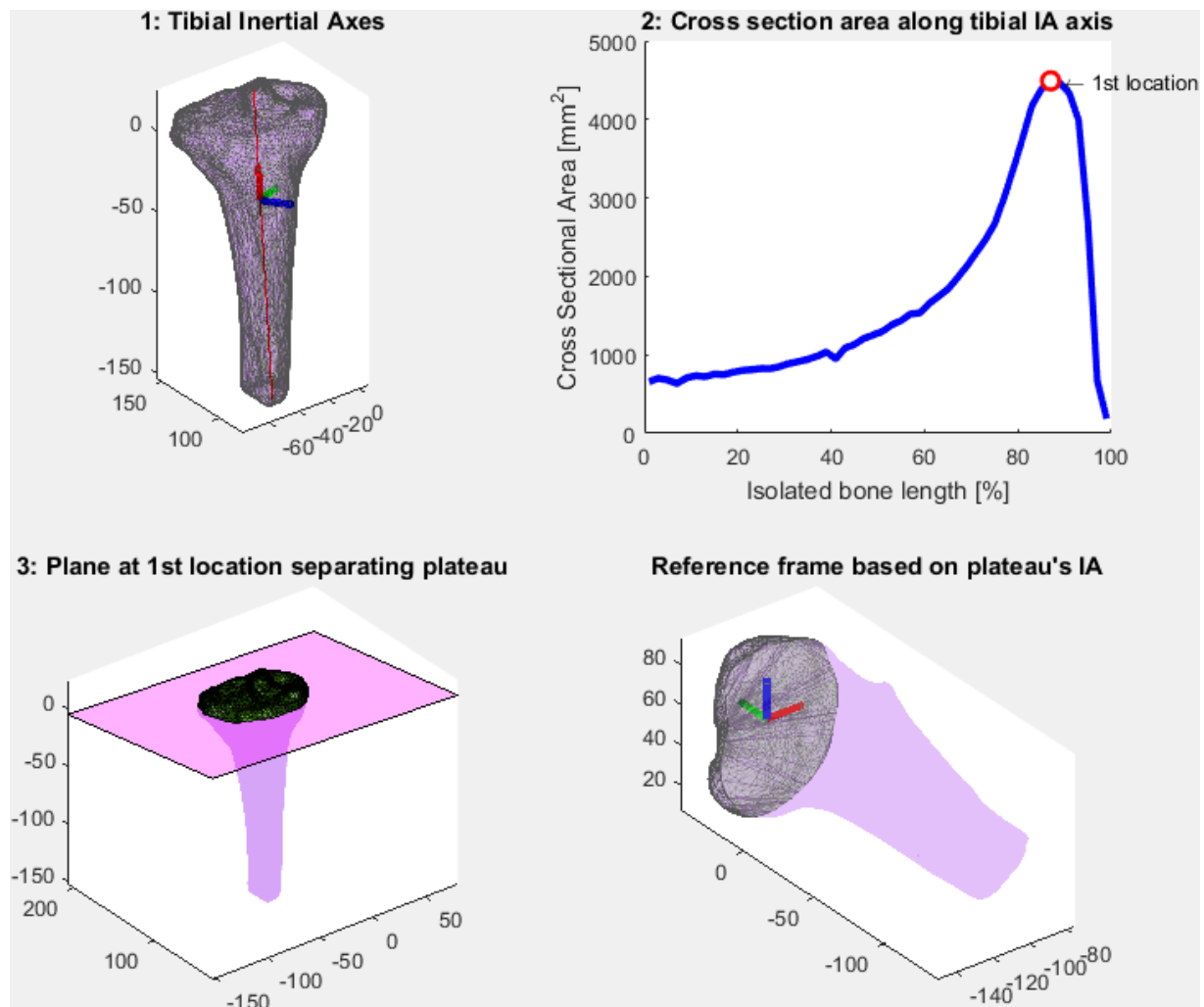


Figure 7: Output of `plotTibiaDiagInfo.m`: Determining the tibial reference frame. The tibial plateau is separated by cutting the tibia perpendicular to the long inertial axis at the widest point (location 1). The inertial axes of the plateau (aligned to the external frame) are used as the reference frame. The axis directions of the last panel are in accordance with the external reference frame. The unlucky viewing angle in the last frame make the resulting frame look bad, but figure 12 with the same axes shows their fine.

Femoral reference frame

Zevenbergens femoral reference frame is based on the long inertial axis of the (distal) diaphysis, and the line connecting the centers of two cylinders fitted to the articulating surfaces of the femur. The process and therefore the exact definition of these lines involves more steps than the patella and tibia combined and is illustrated in 4 figures below.

Remember that our axes are defined as follows:

- Z (ML): the line connecting the centers of two cylinder fits to the articulating surfaces
- X (AP): the normal to the plane containing the Z axis and smallest diaphysis inertial axis (along the length).
- Y (PD): In said plane, and perpendicular to Z. (consequently almost the diaphysis inertial axis)

From this it is obvious that we need to 1) Identify the articulating surfaces and 2) Identify the diaphysis.

To identify the diaphysis, the cross sectional area along the long inertial axis of the whole femur is used (see fig 8). Location 1 is at maximum height, location 2 is at half maximum height, and location 3 is 1.3 times further from the most distal point. A plane perpendicular to the long IA axis direction at location 3 is used to separate the diaphysis.

The condyles are separated using two points and a direction. PT1 is determined by extending the diaphysis long axis from its CM to the distal surface. PT2 is found by taking the plane at location 2 and from the plane's CM extend a line in the direction of the largest femoral IA axis. The condyles are cut with the plane going through PT1 and PT2, and parallel to the femoral second IA axis. A cylinder is fit to the separated condyles. To improve the separating plane for the condyles, the axis of this cylinder is used as a direction to recut the condyles, in conjunction with the PT 1 and 2.

Another cylinder is fit to the condyles, and its axis is extended to the surface to get two points somewhere near the epicondyles. Miranda[3] use the midpoint of these two points as the origin, and the direction of the axis as ML axis.

Abiding to Lianne's idea of fitting cylinder to the articulating surfaces, those surfaces are separated from the condyles based on 4 geometrical measures on the vertices. Figure 11 illustrates these. These measures are

1. The direction of the vertex normal compared to Miranda's ML axis.
2. The distance to the plane separating the condyles from the femur (through PT1 and PT2, and perpendicular to Mirandas ML axis)
3. The distance to the plane separating the condyle from each other (going through PT1 and PT2)
4. The distance to Miranda's ML axis.

All these measures are thresholded to make a binary decision on whether a vertex is an articulating surface vertex or not. Only when all 4 measures designate a vertex as an articulating surface vertex, it is taken as such. Therefore the individual measures are tuned relatively loosely to the dataset. The result is a good segmentation of the articulating surfaces for all tested bones. The values are:

1. The absolute of the cosine of the angle between the vertex normal and Miranda's ML axis should be smaller than 0.6. This results in angle between about 52 and 127 degrees.

2. Anything closer than 0.35 times the furthest vertex from the back-plane is designated as non-articulating surface
3. Anything closer than 0.24 times the furthest vertex from the middle-plane is designated as non-articulating surface.
4. Anything closer than 0.725 times the furthest vertex from Miranda's ML axis after applying the other measures is designated as non-articulating surface.

Next, a cylinder is fit to each articulating surface (least squared to the vertices). The surface centers (center of mass of the faces) are computed and projected on the respective cylinder axis. A line going through both points defines the ML axis (usually almost but not quite coinciding with either cylinder axis). The origin is in the middle between these points. The PD axis is defined as perpendicular to both others and creates a right handed coordinate system.

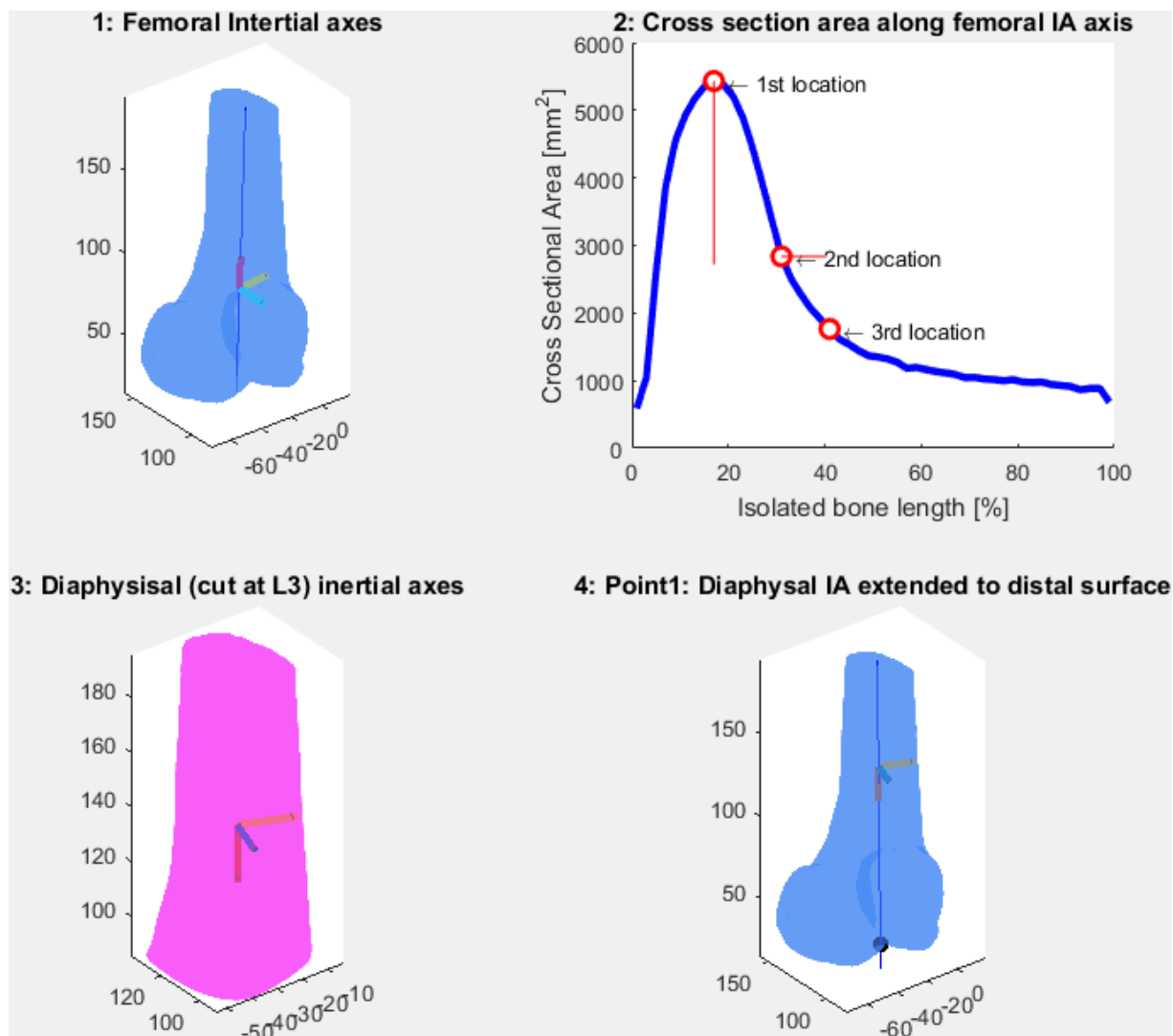
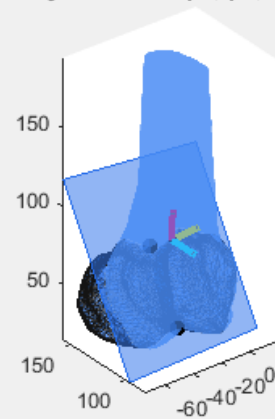
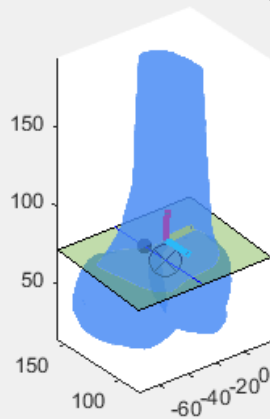
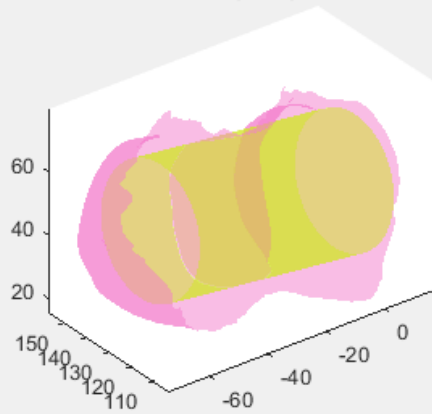


Figure 8: Output of plotFemurDiagInfo 1/4: Crosssectional area and PT1. **1:** Femoral inertial axes. The blue line indicates the cross section area line of panel **2**: Cross section area along the long IA axis of the femur. Locations 1 2 and 3 are respectively peak height, half peak height (on the correct side (towards “tail”)), and 1.3 times more proximal than location 2. **3:** Separated diaphysis with a plane perpendicular to the long inertial axis of the femur (blue line in panel **1**) at location 3. Inertial axes are calculated. **4:** Condyle separating point PT1 is found by extending the diaphysal smallest inertial axis from its center of mass to the distal surface.

5: Point2: Femoral IA extended from plane's(at L2) CoM 6: First condyle cut with p1, p2, and femoral IA



7: First condyle cylinder fit



8: Recut condyles with p1, p2 and first fit's direction

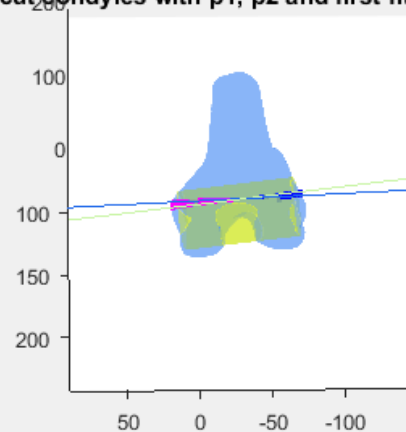


Figure 9: Output of plotFemurDiagInfo 2/4: Neatly separating the condyles. **5:** PT2 is found by finding the posterior intersection of a line from the CM of the plane at location 2 in the direction of the largest femoral IA axis. **6:** The condyles are separated by the plane going through PT1 and PT2, and parallel to the middle femoral IA axis. **7:** A cylinder is fit to the separated condyles. **8:** The axis of this cylinder is used as a direction to recut the condyles, together with PT1 and PT2.

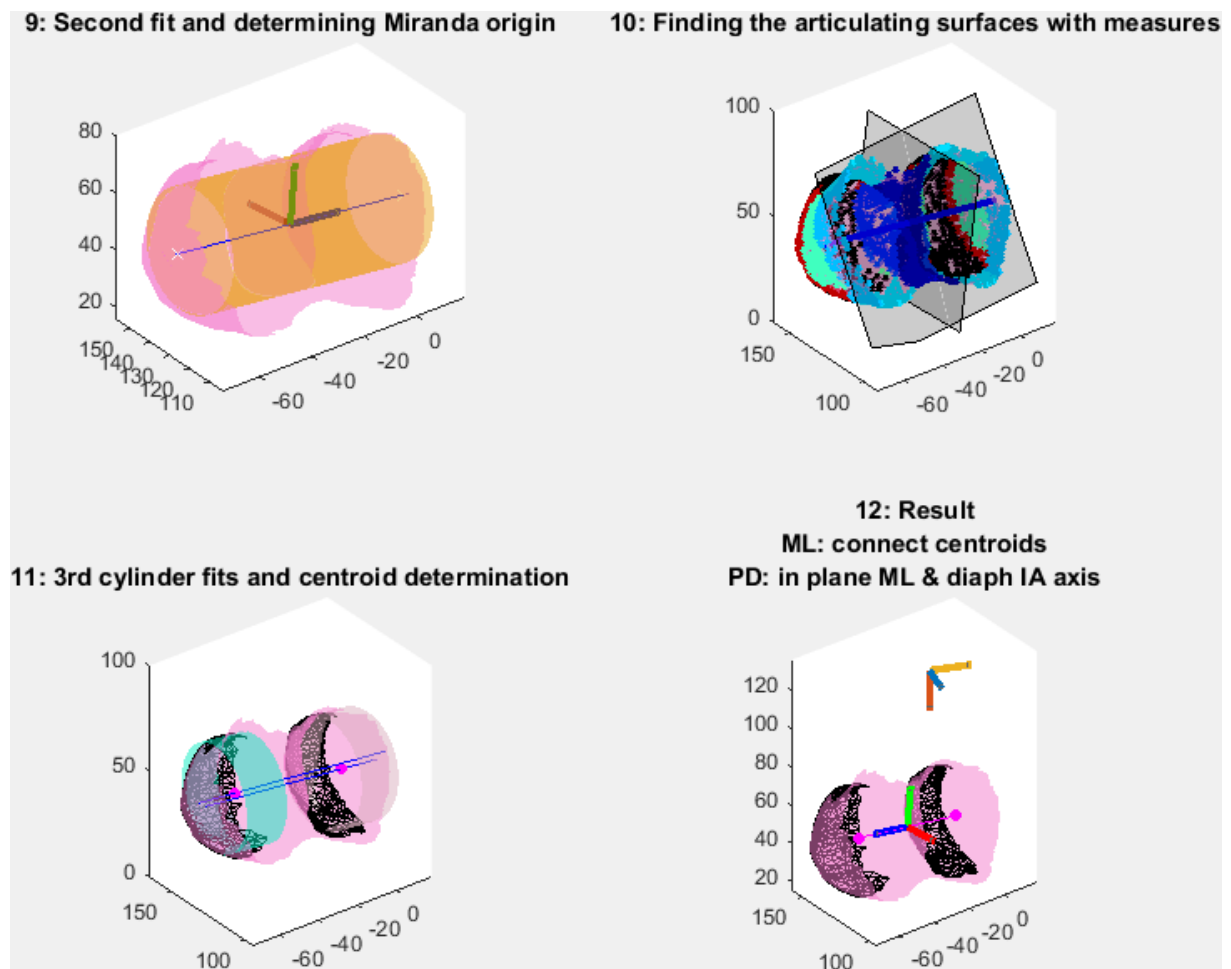


Figure 10: Output of plotFemurDiagInfo 3/4: Finding the ML axis. **9:** Another cylinder is fit to the recut condyles, and its axis is extended to the surface to get two points near the epicondyles. Miranda[3] use the midpoint of these two points as the origin, and the direction of the axis as ML axis. **10:** The articulating surfaces are segmented from the condyles using several measures based on two cutting planes and Miranda's ML axis. See figure 11 for details. **11:** A cylinder is fit to each articulating surface. The articulating-surface-centroid (surface CM) is projected onto its respective cylinder axis (pink dots on blue lines). **12:** Resulting reference frame. The origin is in the middle of the pink dots, the ML axis (blue) is in the direction connecting those dots. The PD axis (green) is almost parallel to the diaphysis long IA axis (red-ish), and perpendicular to the ML axis. The AP axis is perpendicular to both.

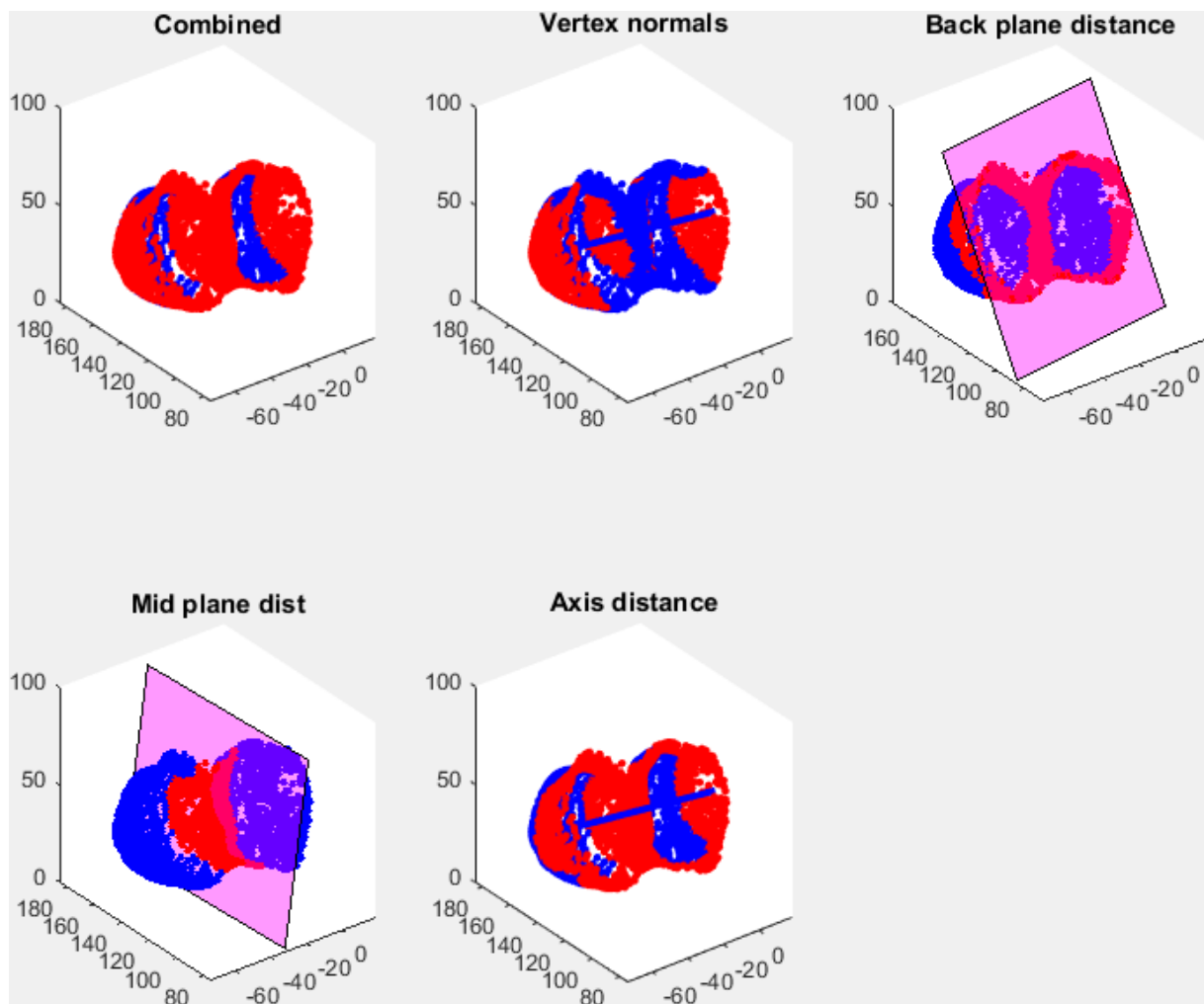


Figure 11: Output of plotFemurDiagInfo 4/4: Segmenting the vertices of the articulating surfaces using geometrical measures. **A)** The measure uses is a combined measure with 4 components, in such a way that each individual measure has to segment the vertex as an articulating surface vertex. Therefore these measures are all tuned relatively loosely, on the 40 bone dataset, making sure to minimize misdesignating of articulating surface vertices. Together they segment the articulating surface really well. **B)** The first measure is the vertex normal measure, comparing miranda's ML axis direction to the vertex normals. If the angle is too small, it's not an articulating surface. **C)** Next the distance to the second plane used to separate the condyles is measured, and to close vertices are identified as non-articulating surface. **D)** The plane going through Pt1 and Pt2, and perpendicular to Miranda's ML axis is used to identify vertices between the condyles as non-articulating surface. Finally the distance to Miranda's ML axis is used. All measures contribute to the final result for all tested bones.

Verification experiment

The reproducibility of the different reference frames has been optimized and verified with 9 datasets (segmented patella and partial tibia and femur). From these experiments an optimized parameter set has been determined and implemented.

Data

The datasets were selected from the 34 available sets in the TLEM-safe dataset + 6 sets from H.Naghbi based on high-median-low values of three measures. These measures are the number of faces, the average face area, and the average angle between neighboring vertex normal (8 of these from the TLEM dataset).

'Ground truth'

In order to vary the input whilst ensuring a sort-of ground-truth, the bones of one dataset are rotated and mirrored in different ways, and the variation in the resulting reference frames is investigated. In a preliminary experiment 118 such transformations were tested, and 12 of those (including the original orientation) were selected to form the final test.

Measure

The results are compared to baseline, which is the result of the original dataset. As quantitative measures, I take the sin of the angle between all 3 axis of the baseline and the data (length of outer product of normalized vectors), as well as the Euclidean distance between the different origins, resulting in 4 values per dataset per bone (down from 6), as well as the runtime. More qualitative figures are provided that show 12 different frames transformed back and overlaid in one image. These confirm the quantitative results. Remember that with the small angles in radians, $\sin(x) \approx x$.

Goal

The goal of these experiments is to quantify the reproducibility of the result and runtime dependence on the parameter values, as well as finding an optimal parameter set, that achieves good reproducibility in reasonable time.

Conclusions

Patella

The inertial axis grid size determines the accuracy of finding the origin, with the mean variation of the same order as the grid size. The AP axis (X) is insensitive to the grid size, and the Y and Z axes do depend on it, but not a lot. Smaller grid sizes result in longer runtimes, with exponential increase behavior for very small values.

Using an optimized parameter set we find the origin with 0.5mm precision, the deviations in the sine of the angles are about 0.05, and it takes about 2 seconds.

Tibia

The grid size is way more important than the step size for a good performance, but also has the biggest effect on runtime.

Using an optimized parameter set we find the origin with 0.5mm precision, the deviations in the sin of the angles are about 0.0002, and it takes about 6 seconds.

Femur

The second inertial axes grid size (of the diaphysis) is much more important than for the first one. Counterintuitively decreasing a grid size (either) does not always increase the reproducibility of the procedure. Tuning the tolerances of the cylinder fitting is important for finding the right trade-off between runtime and accuracy. Skipping vertices (FirstFitUseEvery>2) for the first cylinder fit is a viable method for increasing performance, but appropriate tolerances are more important.

Using an optimized parameter set we find the origin with 0.05mm precision, the deviations in the sin of the angles are about 0.001, and it takes about 10 seconds.

Qualitative results

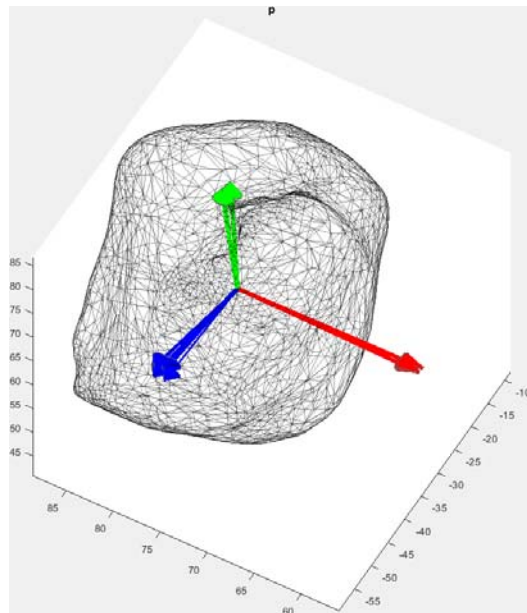
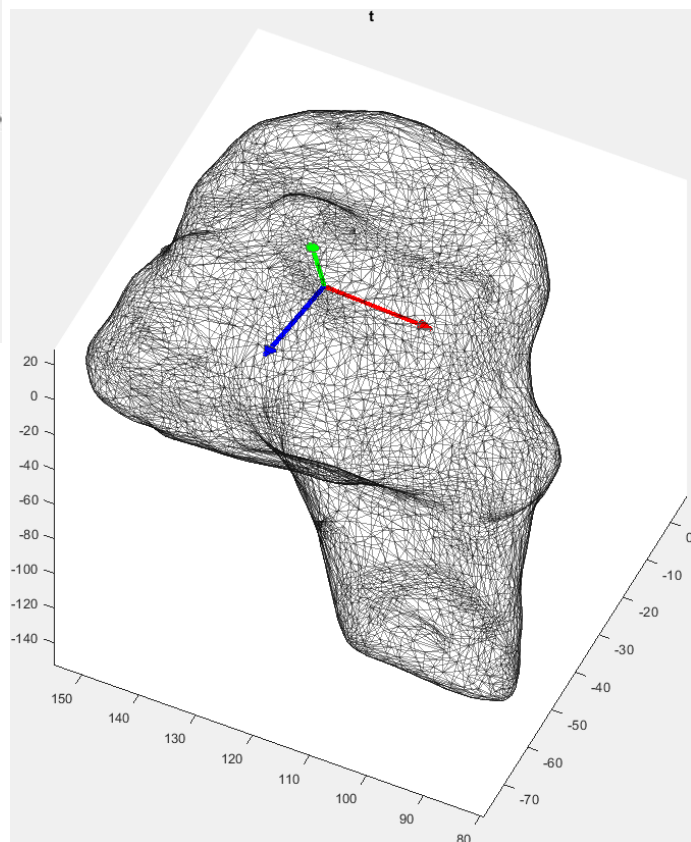


Fig 13: All 12 generated axes for the tibia, transformed back to the original coordinate system. These results are found using an optimized parameter set. This is quite good.

Fig 12: All 12 reference frames from the rotated and mirrored version of the same bone, transformed back to the original coordinate system. These results are found using an optimized parameter set. Unlike for the other bones, no deviations are visible when the reference axes are 20mm long.



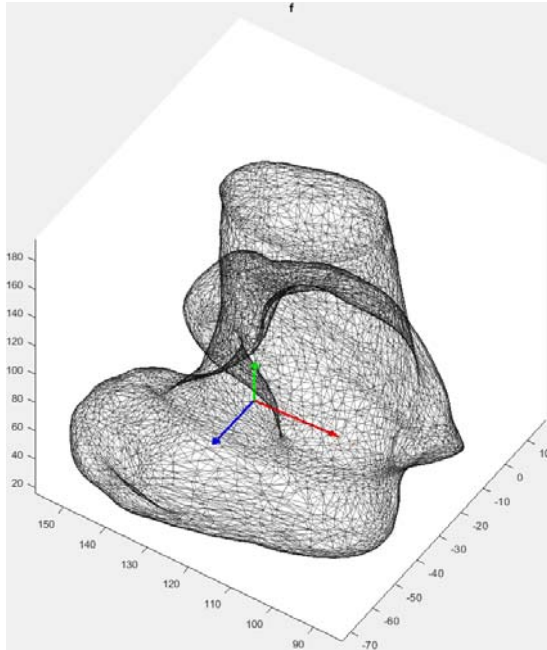


Fig 14: All 12 femoral reference frames generated in the verification experiments, transformed back to the original coordinate system. These results are found using an optimized parameter set. For the femur, the correspondence is magnificent.

Open questions

It is to be determined whether a small enough grid size suffices, or a new method has to be developed, for instance on the basis of surface integrals over the triangle mesh and applying Stokes' theorem.

Although I think small grid is fine².

How sensitive is the procedure to mesh quality,

Both in runtime and in accuracy. Can we improve performance with specific meshes? i.e. high detailed around inferior pole? Can we define limits / generation procedures that are favorable for the result? Is it important that these are followed to get a reproducible result? A usefull experiment would be to have different people process the same binary map to a surface mesh, (possibly in more than 1 way per person), and examine the consistency of the result.

How problematic is it that we are not precutting the Tibia?

Unlike the femur, the tibia is not cut to a determined length before processing. This is done because the tibia is way more straight and we do not use the tibial shafts inertial axes to determine the PD axis. This all makes sense, but the effect of tibial length on the resulting frame have not been quantified.

² No pun intended

Changes from the Lianne code

The inferior pole finding algorithm has been introduced.

The frame estimate is introduced, enabling doing the manual parts before computations instead of during, as well as eliminate all manual steps in certain situations.

Reference length has been introduced

The quality of the segmentation of the articulating surfaces has been improved by a lot.

Fixed issues that prevented the code from working in left knees.

The structure of the program has been reworked extensively, renaming, moving and combining functions and procedures to produce more understandable and easier to work with code.

The inertial axes finding algorithm was fairly broken, in more than one way. The problems mainly arose where sections of bones were separated and then analyzed. This was because the inertial axes finding algorithm uses a flood filling procedure to determine the point-mass-distribution of the bone. Closing the holes left by separating was flawed, and failed to close the holes at one end, and added faces to an already closed surface somewhere else. These problems have all been fixed:

- The cutting procedure was update to move vertices of faces around the cutting edge towards the edge, instead of cutting them and removing the faces.
- This enabled cleanly closing the surfaces using Delaunay triangulation on the vertices on the edge
- To remove faces in exterior (concave (e.g. between the articulating surfaces)) areas generated by the Delaunay triangulation, an iterative face pruning procedure was implemented

Fixed a clear bug in the patella script caused errors in the ML and PD axes; due to a typo, the PD axis was not drawn from the CM (like (CM_x, CM_y, CM_z)) to the inferior pole, but from another point given by (CM_x, CM_y, CM_y) .

All parameters have been retuned on the basis of 9 datasets to increase reproducibility.

References

- [1] W.J. Zevenbergen. Improved anatomical coordinate system of the distal femur based on 3D bone geometry & evaluation of the inter- and intra- observer variability of the knee ligament attachment sites. (Unpublished Master's thesis) University of Twente
- [2] N.T. Kruis. 3D laxity assessment in ACL-patients (Unpublished Master's thesis) TU Eindhoven
- [3] Grood ES, Suntay WJ. A joint coordinate system for the clinical description of three-dimensional motions: application to the knee. J Biomech Eng. 1983 May;105(2):136-44. PubMed PMID: 6865355.
- [4] Miranda, D. L., M. J. Rainbow, et al. (2010). "Automatic determination of anatomical coordinate systems for three-dimensional bone models of the isolated human knee." J Biomech 43(8): 1623-1626.
- [5] Sharma 2012; Radiological method for measuring patellofemoral tracking and tibiofemoral kinematics before and after total knee replacement
- [6] Ho 2012; Computed tomography analysis of knee pose and geometry before and after total knee arthroplasty